

Alexander Tereshin

AI Research Engineer

Agentic Systems · RAG · Production LLM

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AI Research Engineer building **agentic systems** that hold up in production. Architected a corporate **multi-agent AI platform** at a major bank, where autonomous agents generated product hypotheses with **67% accepted** by product teams; now developing **agentic VLM-based RAG** that outperforms published SOTA baselines on financial-document QA. Grounded in **classical ML** and **NLP**: forecasting, production classification pipelines, and large-scale A/B experimentation in e-commerce.

EXPERIENCE

NLP Researcher AIRI

May 2026 – Present

- Designing an **agentic VLM-based RAG** system for text-and-table document QA over real-world **financial reports**, outperforming published SOTA baselines in numerical-reasoning accuracy (**Number Match**) on **T²-RAGBench**; first-author conference paper in preparation.

Teaching and Research Assistant Higher School of Economics (HSE University)

Sep 2025 – Present

- Teaching assistant for graduate **Basic & Advanced NLP** courses (300+ students); designed curriculum and examined 30+ students on topics from **Transformer internals** to **Multi-Agent Systems**;
- Supervised low-level LLM implementation - manual **Qwen2.5** architecture (**RoPE**, **RMSNorm**, **SwiGLU**) for speculative decoding, plus **PEFT/QLoRA** fine-tuning and **GPTQ** quantization.

Machine Learning Engineer Raiffeisenbank

Jun 2024 – May 2026

- Architected and delivered a corporate **multi-agent AI platform** where autonomous agents explore user data to **generate product hypotheses**, with **67% accepted by product teams** for further testing. Equipped agents with an **Agentic RAG** tool (**FastMCP**) for banking-domain context and a **Zero-Touch QA agent (Ragas)** for automated evaluation, served by an internal **vLLM** stack — **GLM** for agent reasoning and **Qwen-2.5-Coder** for code generation – in isolated Micro Sandboxes;
- Engineered an async **large-scale** data labeling and experimentation pipeline (**Litestar**, **FastStream**, **RabbitMQ**), scaling throughput **10x** (to **2K labels/day**) at **92% accuracy** via **Qwen2.5 Few-Shot CoT**;
- Designed and deployed 4 production batch pipelines, orchestrating **PySpark** jobs on **HDFS** via **Airflow** to serve a **LightGBM** model (**81% F1**); built full-stack observability and drift monitoring (**Evidently AI**, **Prometheus**, **Grafana**) to detect signal degradation and non-stationary behavior in production;
- Built a Hybrid RAG system (**Milvus**, **BM25** + **BGE-M3**) automating **80%** of basic HR policy responses, cutting resolution time by **99%** (hours to seconds) at **80 ms retrieval latency**.

Data Scientist — Retail & E-commerce

Jan 2023 – Jun 2024

Sportmaster (Jan–Jun 2023) · VseInstrumenti.ru (Jun 2023 – Feb 2024) · Leroy Merlin (Feb–Jun 2024)

- Led Data Science for three core products on a **12M+ MAU** platform (VseInstrumenti.ru), running **40+ online A/B experiments** with advanced variance reduction (**Stratification**, **Delta Method**) for robust statistical inference, directly driving a **\$25K** monthly revenue increase;
- Built churn forecasting with **SARIMA** and interpreted a **CatBoost** model via **SHAP** to surface churn drivers and retention insights for sales and product teams; delivered a production ML classification pipeline for warehouse safety-risk detection.

Career Transition into Machine Learning

Feb 2022 – Jan 2023

Full-time self-study

- Full-time transition into ML: statistics, Python, and core algorithms, leading to admission to the HSE Master's program in Machine Learning and High Load Systems.

EDUCATION

Higher School of Economics (HSE University)

Master's degree in Machine Learning and High Load Systems

Sep 2023 – Jun 2025

GPA: 8.29/10

Saint Petersburg State University of Civil Aviation

Bachelor's degree in Maintenance of Aircraft and Engines

Sep 2016 – Jun 2021

EARLIER EXPERIENCE

Aviation Engineering & Operations

2018 – 2022

- Progressed from Airframe & Powerplant Engineer to **Lead Engineer** in business aviation and then **Production Shift Manager**, leading cross-functional teams of **60+ people** across 30+ safety-critical Boeing / Airbus maintenance projects; managed data-driven resource allocation and AOG (aircraft-on-ground) recovery under strict reliability constraints.

SKILLS

Programming & ML: Python, PyTorch, Transformers, scikit-learn, LightGBM, CatBoost

Agentic & LLM: Agentic Workflows, LangChain, LangGraph, Pydantic AI, MCP, RAG, PEFT/QLoRA, GPTQ

Evaluation & Statistics: Experimental Design, A/B Testing, Variance Reduction, Drift Monitoring, Ragas

MLOps & Orchestration: Airflow, Kubernetes, Docker, CI/CD, MLflow, vLLM, Evidently AI, Prometheus, Grafana

Data & Databases: PySpark, SQL, Hadoop, Vector DBs (Milvus), MongoDB, AWS S3

Backend & Tools: FastAPI, Litestar, Kafka, RabbitMQ, Redis, Bash, Linux, Git, Pytest

Languages: English (B2), Russian (Native)